Evaluation Generated Clinically Meaningful Clusters

(Evaluation guidelines)

Outline

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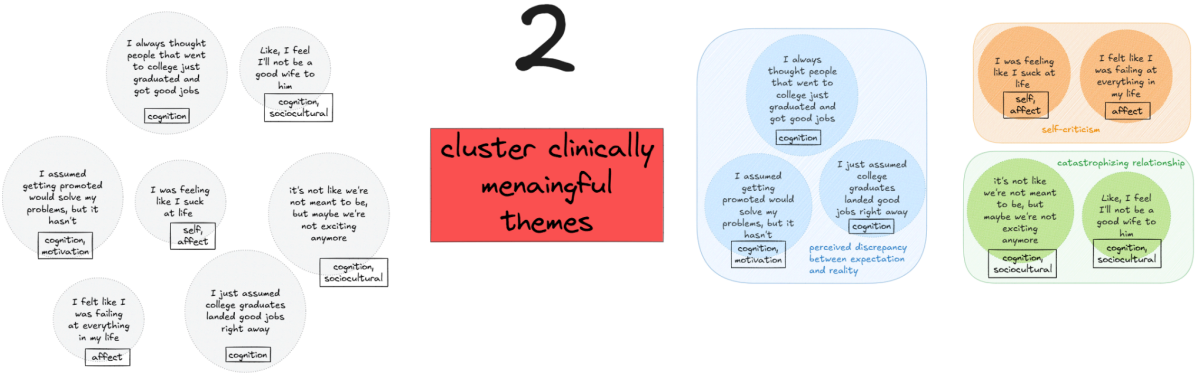
1. Introduction

Why is the Evaluation of the Cluster Important?

Mental health issues represent a major global public health challenge, and while numerous therapeutic strategies are available, current clinical psychology often relies on standardized protocols that may fail to fully address the unique needs of individual patients. Processed-Based Therapy (PBT) offers a novel approach by integrating multiple effective treatment strategies tailored to individual needs, guided by the Extended Evolutionary Meta-Model (EEMM). The EEMM serves as an assessment tool (graph network) that transcends specific theoretical or intervention frameworks (e.g., CBT, ACT), and instead outlines areas of coverage and principles of change that clinical theories must address when considering humans as evolving systems.

However, constructing EEMM networks requires therapists to manually annotate therapy sessions to identify key psychological processes and their interrelationships, posing significant scalability challenges. This project seeks to overcome these limitations by automating the entire process of building EEMM (aka client networks) from annotated therapy sessions. In particular, this thesis is set up to identify and consolidate redundant processes (nodes in the client network), cluster processes that share highly similar clinical themes, and predict explainable relationships between these themed clusters, thereby enhancing the efficiency and scalability of PBT implementation.

We are currently in work package two, where we aim to generate clinically meaningful themes through the use of large language models (no connection between themes, this will be the next Work Package). By evaluating the clusters generated from the mini-session, you will help to determine how effectively the model groups related processes and generates clinically meaningful themes for each cluster. This evaluation will provide valuable insight into the performance and overall usefulness of the model, which will be used to produce the EEMM.



2. Overview of the evaluation

Cluster

The cluster you are going to evaluate was generated by a large language model using input from a single mini-session from a patient. This mini-session includes pre-labeled processes and a prompt that guides the model in producing the cluster.

Provided Material

To evaluate the generated Cluster from a mini-session, you will be provided with an Excel spreadsheet along with the dialogue from the mini-session. This will give you an overall context of the mini-session's content. The dialogues are also labeled with the corresponding processes.

Key Definitions

- **Large Language Model (LLM)**: A LLM is an advanced artificial intelligence system trained on vast amounts of text data to understand, generate and respond to human language in a meaningful way.

- **Process:** A process is a series of connected actions or changes that happen over time and are influenced by the situation. In psychology, processes explain how and why problems develop and change, helping guide personalized treatment.

- **Dimension:** A broad category representing different aspects of psychological function (e.g., cognitive, emotional, behavioral), within which processes operate.

- **Cluster:** A cluster is a group of related nodes (processes with labels) that the large language model has grouped together based on its interpretation of common clinical themes. Each cluster reflects a specific clinical theme derived from these nodes.

- **Frequency**: Each process has a frequency, indicating how often a semantically similar process occurred in the dialogue. A higher frequency suggests that the process itself holds greater significance.

- **Dialogue**: The provided dialogue is a transcript from a mini-session, specifically the one you are going to evaluate. It includes processes labeled with dimension and frequency.

- **Generated Cluster Name (Clinically Theme)**: This is the name assigned to a Cluster. The name reflects the overall clinical theme or concept that unifies these nodes based on their content, making the cluster's subject/topic/idea clear.

- **Cluster Set:** The Cluster Set represents the complete collection of clusters generated by the model. Each cluster comprises a generated clinical theme and the utterances or processes assigned to that theme based on their shared content or clinical context. The individual clusters are independent and are not necessarily connected to one another.

- **Node (Process, Frequency, and Dimension):** A Node represents a specific process identified during a patient's session (mini session), paired with a label that categorizes this process and frequency.

- **Evaluation Criteria Network Level**: This level evaluates the overall structure and quality of the cluster set.

- **Evaluation Criteria Cluster Level**: This level evaluates the quality and relevance of individual clusters, including the Generated Cluster Name.

- **Evaluation Criteria General Assessment**: This level assesses the combination of Cluster and Network Level

Evaluation Criteria Network Level: Definition and Example

Criteria	Definition
Specificity	Specificity measures whether the network identifies only relevant processes/nodes, avoiding overly broad or unrelated information.
Novelty	Novelty evaluates whether the network provides new, unexpected, or useful insights beyond existing knowledge of the client/therapy session.
Completeness	Completeness measures whether all processes have been incorporated into the network. (Automatic, not evaluated by you)
Coverage	Coverage measures whether the network covers all relevant clinical topics in the current mini-session, ensuring that nothing important is missing.
Propose Clinically Theme (Optional)	The evaluator can suggest an alternative clinical theme/title for a cluster if the generated theme does not fully capture the meaning/theme of the nodes. This helps us to refine the model's output by incorporating clinical expertise and improving the relevance and accuracy of the themes.
Redundancy	Redundancy assesses whether the network contains duplicate or overlapping clinical themes, reducing efficiency and clarity.
Clinical Relevance	Assesses whether the identified clinical theme is meaningful to client well-being/functioning/goals.
Comments	Here you can add some comments about the cluster set itself

Criteria Specificity:

Criteria	Definition
Specificity	Specificity measures whether the network identifies only relevant processes/nodes, avoiding overly broad or unrelated information.

Value	Example
1 Too Broad	The network clusters sentences like *"I feel like a failure," "I'm always stressed,"* and *"I don't sleep well"* under a broad theme like "Mental Health Issues." This is too general to provide useful clinical insights.
2 Moderately Specific	The network clusters sentences like *"I'm overwhelmed by work"* and *"I can't relax because of my job"* under a theme called "Work Stress." This is somewhat specific but could be refined further.
3 Highly Specific	The network clusters sentences like *"I worry constantly about meeting deadlines," "I feel paralyzed by fear before presentations,"* and *"I avoid talking to my boss"* under the theme "Work-Related Anxiety." This is highly specific and clinically useful.

Criteria Novelty:

Criteria	Definition
Novelty	Novelty evaluates whether the network provides new, unexpected, or useful insights beyond existing knowledge of the client/therapy session.

Value	Example
1 None	The clusters simply reflect common categories like "Stress" and "Anxiety" without highlighting new insights.
2 Moderately Insightful	The network groups *"I can't stop worrying about my performance"* with *"I'm afraid of being judged,"* suggesting a common theme of "Performance Anxiety," which is useful but expected.
3 Highly Novel	The network groups *"I feel like I'm failing as a parent"*

	with *“I feel guilty when I relax,”* revealing a novel theme of “Parental Perfectionism,” providing a previously unknown clinical insight.

Criteria Coverage:

Criteria	Definition
Coverage	Coverage measures whether the network covers all relevant clinical topics in the current mini-session, ensuring that nothing important is missing.

Value	Example
1 Incomplete	The network captures themes like “Work Stress” but misses themes like “Family Conflict” and “Self-Worth Issues,” even though the patient mentioned statements like *“I feel like a bad mother”* or *“My partner and I argue constantly.”*
2 Partially Complete	The network includes themes like “Work Stress” and “Anxiety” but misses “Low Self-Esteem,” despite the patient saying things like *“I never feel good enough”* or *“I think I’m a failure.”*
3 Fully Complete	The network captures all relevant themes, such as “Work Stress,” “Family Conflict,” “Self-Worth Issues,” and “Anxiety,” covering statements like *“I feel like a failure,” “I can’t balance work and family,”* and *“I’m always anxious about my performance.”*

Criteria Redundancy:

Criteria	Definition
Redundancy	Redundancy assesses whether the network contains duplicate or overlapping clinical themes, reducing efficiency and clarity.

Value	Example
1 Highly Redundant	The network has separate clusters for *"I feel like a failure"* and *"I never feel good enough,"* both under themes like "Low Self-Esteem" and "Insecurity," which overlap significantly
2 Moderately Redundant	The network clusters *"I can't handle my job"* under "Work Stress" and *"I'm overwhelmed by my workload"* under "Burnout," which share some overlap but have subtle differences.
3 Minimal Redundancy	Each cluster is distinct: *"I can't relax because of work"* under "Work Stress" and *"I don't feel competent in my role"* under "Low Self-Esteem." No themes are duplicated, but the same node is allowed to be included in multiple clusters.

Criteria Clinical Relevance:

Criteria	Definition
Clinical Relevance	Assesses whether the identified clinical theme is meaningful to client well-being/functioning/goals.

Value	Example
1 Clinically Insignificant	The Cluster set shows clusters with vague or irrelevant themes, such as grouping "mood swings" and "sleep patterns" without any clear clinical connection or practical use.

	The cluster set includes clusters like *“I feel tired”* and *“I don’t enjoy things”* under a vague theme like “General Discomfort,” which isn’t useful for clinical diagnosis.
2 Partially Meaningful	The cluster set contains some useful clinical themes, like “anxiety disorders,” but are too broad or lack focus, making it difficult to apply them directly in clinical decision-making. The clusterset *“I can’t sleep”* and *“I’m constantly worried”* under the theme “Stress,” which is somewhat helpful but lacks specificity.
3 Clinically Significant	The network identifies clear, actionable clinical themes, such as “excessive worry about health,” and “hypervigilance for threat” providing meaningful insights that can support diagnosis or treatment planning. The cluster set *“I worry excessively about my kids,” “I feel like a bad mother,”* and *“I’m afraid of making mistakes with parenting”* under the theme U“Parental Anxiety,” which is clinically actionable.

Evaluation Criteria Cluster Level: Definition and Example

Criteria	Explanation
Intrusiveness	Intrusiveness identifies whether (if any) node in the cluster does not fit or is irrelevant to the cluster overall theme **Focus:** Identifying specific nodes that do not fit within a cluster
Comments	Here you can add some comments about each Cluster

Criteria Intrusiveness:

Criteria	Definition
Intrusiveness	Intrusiveness identifies whether (if any) node in the cluster does not fit or is irrelevant to the cluster overall theme **Focus:** Identifying specific nodes that do not fit within a cluster

Value	Example
1 Multiple Intrusivenesss	The cluster is labeled **"Work Stress"** but contains nodes like *"I feel like a bad parent"* and *"I don't enjoy social events,"* which do not fit.
2 Few Intrusivenesss	The cluster is labeled **"Anxiety Issues"** and contains mostly relevant nodes, like *"I worry about everything,"* but includes one out-of-place node, *"I feel worthless."*
3 All Relevant	The cluster is labeled **"Work Burnout"** and contains only fitting nodes, such as *"I feel exhausted by my job," "I can't keep up with my workload,"* and *"I dread going to work."*

Evaluation Criteria General Assessment: Definition and Example

Criteria	Definition
Usefulness	Usefulness evaluates whether the network offers valuable insights or supports clinical decision making effectively
Comments	Here you can add some comments about the overall Performance of Network and Clusters

Criteria Usefulness:

Value	Example
1 Limited Insight	The network and cluster provide very general or surface-level insights, making it difficult to derive meaningful clinical conclusions. Example: Clusters statements like *"I feel tired," "I can't concentrate,"* and *"I feel sad"* under a vague theme like "General Mental Health Issues." The themes are too broad and do not help clinicians identify specific problems or guide treatment decisions.
2 Some Insight	The network and clusters offer some useful information, but the insights are incomplete or lack depth, limiting their clinical usefulness Example: The Network identifies clusters such as "Work Stress" and "Family Issues". While these themes are helpful, they miss deeper insights or connections. For instance, a cluster on "Low Self-Esteem" could have been more detailed.
3 Valuable Insight	The network and clusters provide detailed, specific, and actionable insights that effectively support clinical decision-making. Example: The network identifies distinct and clinically useful themes like "Parental Anxiety," "Work-Related Burnout," and "Low Self-Worth." Each cluster contains statements like *"I feel guilty when I relax"* (Parental Anxiety), *"I dread going to work"* (Work Burnout), and *"I never feel good enough"* (Low Self-Worth). These

	insights help clinicians identify targeted interventions and understand the patient's unique struggles.

Annotation steps:

1. Read the provided dialogue from the current mini-session.
2. Read all the processes of the mini-session and the generated clinically theme
3. Evaluate the criteria at the Network Level **in order**
4. Evaluate the criteria at Cluster Level at Cluster Level **in order**
5. Evaluate the criteria at General Assessment

Evaluation steps in details

Before starting the evaluation, you will open the evaluation sheet that has been assigned to you.

The first two rows of the evaluation sheet are reserved for the heading, **do not change them!**

	A	B	C	D	E	F	G
1		Node					
2	**Generated Cluster Name (Clinnically Theme)**	**Process**	**Frequency**	**Dimension**	**Specificity:** Is the network too broad or sufficently specific to be clinically usefull?	**Novelty:** Does the network reveal any new, unexpected, or useful insights?	**Coverag** network clinicly t

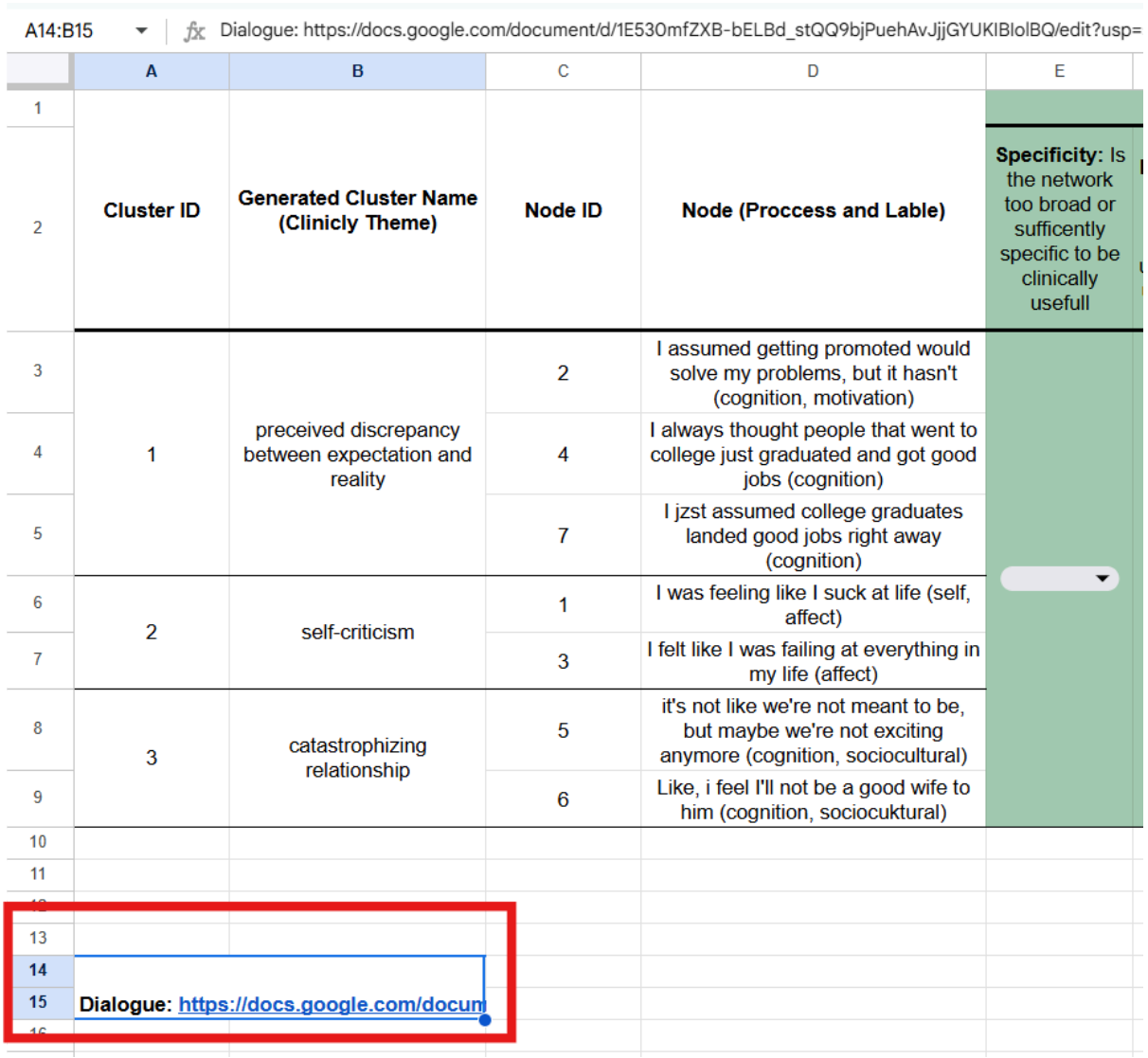
J	K	L	M	N	O	P
		Evaluation Criteria Cluster Level			**Evalaution Criteria General Assessment**	
Clinical Relevance: Is the identified clinicly theme meaningful?	Comments	**Consistency:** Is the theme consistent with the content?	**Indruder:** Is there an intruder node in the Cluster	Comments	**Usefulness:** Can this cluster provide insight or support clinical decision-making?	Comments

The headings are colour coded. Colourless means that this column is the given input and should **not be changed!**
Green represents the network level assessment, **yellow** represents the cluster level assessment, and **purple** represents the overall assessment. The Evaluation criteria in each of those sections is described above (Chapter 2).

Steps:

1. Read the provided dialogue from the current mini-session

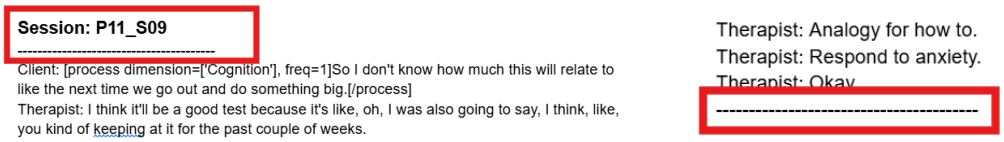
Each dialogue can be found at the bottom of the evaluation sheet. Click on the link to go to the text document



A14:B15 | Dialogue: https://docs.google.com/document/d/1E530mfZXB-bELBd_stQQ9bjPuehAvJjjGYUKIBIolBQ/edit?usp=

	A	B	C	D	E
1					
2	**Cluster ID**	**Generated Cluster Name (Clinicly Theme)**	**Node ID**	**Node (Proccess and Lable)**	**Specificity:** Is the network too broad or sufficently specific to be clinically usefull
3	1	preceived discrepancy between expectation and reality	2	I assumed getting promoted would solve my problems, but it hasn't (cognition, motivation)	
4			4	I always thought people that went to college just graduated and got good jobs (cognition)	
5			7	I jzst assumed college graduates landed good jobs right away (cognition)	
6	2	self-criticism	1	I was feeling like I suck at life (self, affect)	
7			3	I felt like I was failing at everything in my life (affect)	
8	3	catastrophizing relationship	5	it's not like we're not meant to be, but maybe we're not exciting anymore (cognition, sociocultural)	
9			6	Like, i feel I'll not be a good wife to him (cognition, sociocuktural)	
10					
11					
13					
14					
15	**Dialogue:** https://docs.google.com/docum				

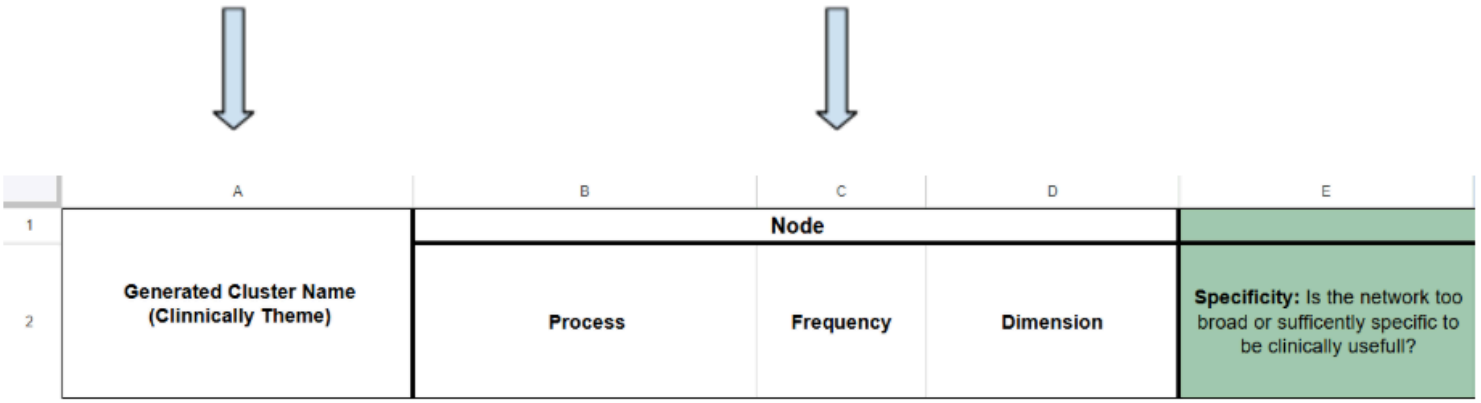
You will find all the listed dialogs assigned to you in this text document.
Each dialogue starts with the patient number and session, followed by dashes. You will find these dashes at the end of each dialogue



Find the dialogues you are currently evaluating and read through them. This will give you a general idea of what this mini-session is about. To find out which mini-session you are currently evaluating, look at the sheet name at the bottom.

2. Read the processes of the mini-session and the generated clinically theme

After reading the dialogue, read the colourless section where the nodes (process, frequency, dimension) and the generated cluster name are written to be able to evaluate the network and the clusters.



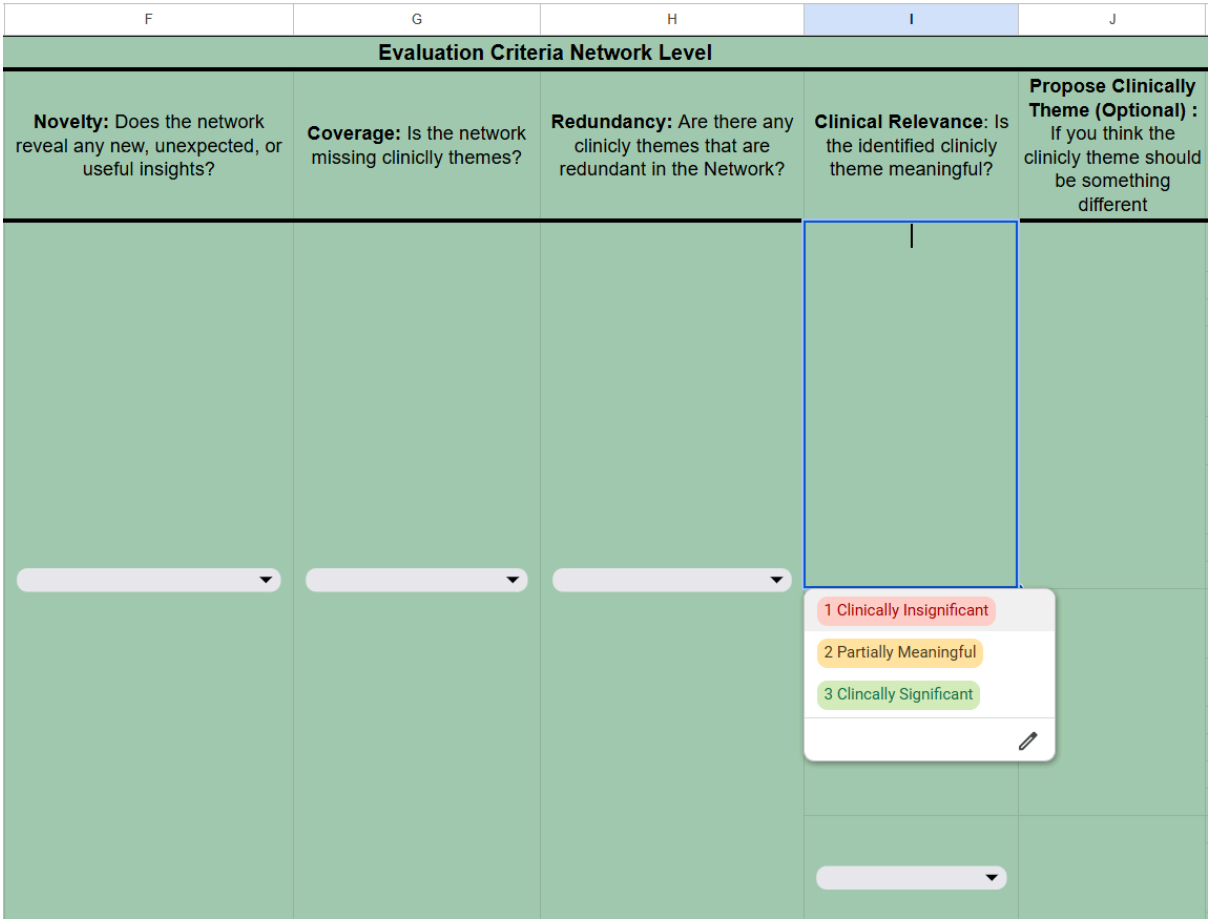
Those are in the columns **A** (Generated Cluster Name/Clinically Theme) and **B-D** (Node with Processes, Frequency and Dimension).

3. Evaluate the criteria at the Network Level in order

Once you have gained an overview of the network, the next step is to evaluate the network level using the criteria provided **in the given order**. These criteria are presented in columns **E** through **J**. In addition, a question is posed for each criterion in order to obtain a more comprehensive understanding.

E	F	G	H	I	J	K
Evaluation Criteria Network Level						
Specificity: Is the network too broad or sufficently specific to be clinically usefull?	**Novelty:** Does the network reveal any new, unexpected, or useful insights?	**Coverage:** Is the network missing clinicly themes?	**Redundancy:** Are there any clinicly themes that are redundant in the Network?	**Clinical Relevance**: Is the identified clinicly theme meaningful?	**Propose Clinically Theme (Optional) :** If you think the clinicly theme should be something different	Comments

Each of the criteria is accompanied by a drop down, each of which is assigned a value between 1 and 3 (plus a few words definition of the number). The second chapter provides a detailed explanation of the criteria, as well as illustrative examples.

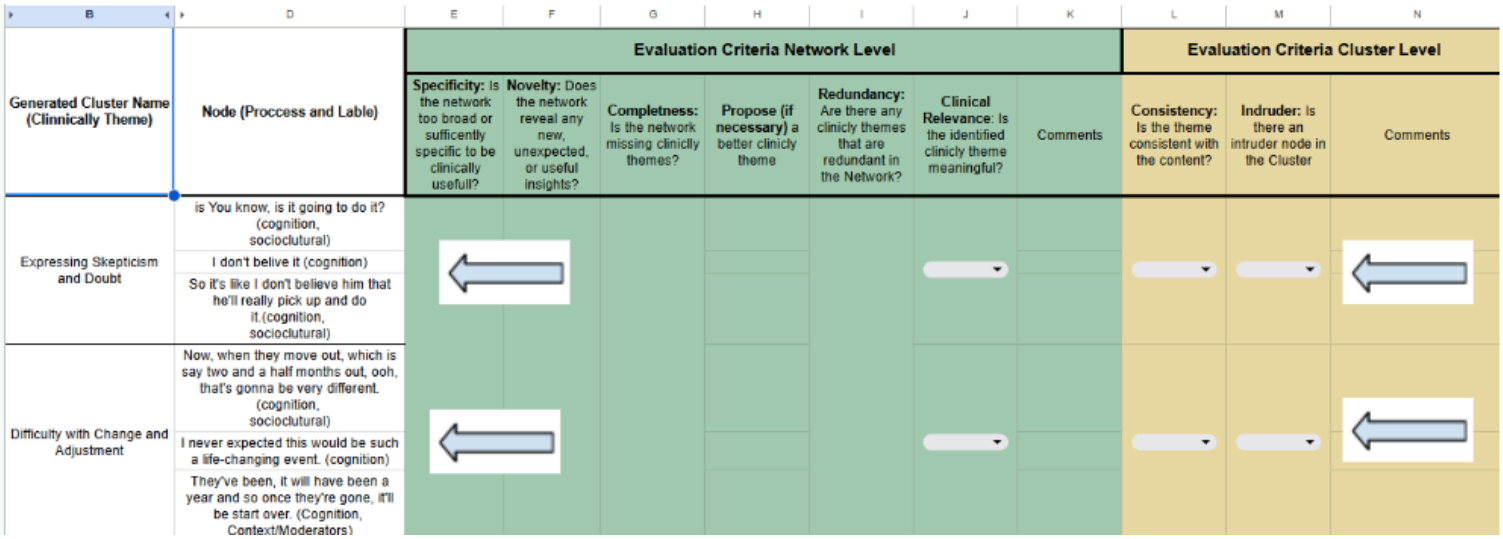


While criteria such as specificity (Column E), novelty (Column F), coverage (Column G) and redundancy (Column I) are common to all clusters, clinical relevance (Column J) is specified for each cluster.

In addition, you also have the possibility to make some comments in column K for the evaluation of the Network Level.

4. Evaluate the criteria at Cluster Level at Cluster Level in order

Once the network has been evaluated, you will evaluate each cluster in the network, which will be shown in yellow (Evaluation Criteria Cluster Level). Columns from L trough M

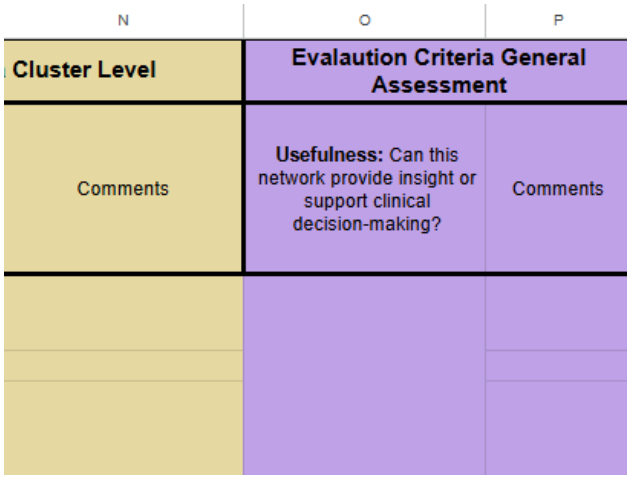


B	D	E	F	G	H	I	J	K	L	M	N
		Evaluation Criteria Network Level							Evaluation Criteria Cluster Level		
Generated Cluster Name (Clinnically Theme)	Node (Proccess and Lable)	**Specificity:** Is the network too broad or sufficently specific to be clinically usefull?	**Novelty:** Does the network reveal any new, unexpected, or useful insights?	**Completness:** Is the network missing clinicly themes?	**Propose (if necessary)** a better clinicly theme	**Redundancy:** Are there any clinicly themes that are redundant in the Network?	**Clinical Relevance:** Is the identified clinicly theme meaningful?	Comments	**Consistency:** Is the theme consistent with the content?	**Indruder:** Is there an intruder node in the Cluster	Comments
Expressing Skepticism and Doubt	is You know, is it going to do it? (cognition, socioclutural)										
	I don't belive it (cognition)										
	So it's like I don't believe him that he'll really pick up and do it.(cognition, socioclutural)										
Difficulty with Change and Adjustment	Now, when they move out, which is say two and a half months out, ooh, that's gonna be very different. (cognition, socioclutural)										
	I never expected this would be such a life-changing event. (cognition)										
	They've been, it will have been a year and so once they're gone, it'll be start over. (Cognition, Context/Moderators)										

This means that you must select a number between 1-3 for **each cluster** in the network in the drop down menu. Additionally like before you have also the possibility to add some Comments into this Category in Column N.

5. Evaluate the criteria at General Assessment

This step is the final part of the evaluation. In this category, which is coded as purple, you will evaluate the network and the cluster together.



N	O	P
Cluster Level	**Evalaution Criteria General Assessment**	
Comments	**Usefulness:** Can this network provide insight or support clinical decision-making?	Comments

Again there is a drop down menu in column O where you can select a score from 1 to 3. There is also a comment section in column P.